School of Mathematics
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Integer and Combinatorial Optimization

Homework Assignment 2

March 12, 2026

Instructions

1. For this assignment, you will have to write a report and produce some code.
2. You should attempt all questions.
3. The total marks for this assignment are: 100.
4. Explain in your report what you have been doing in a concise and reproducible way. Write full sentences, not just the final results. The style in which you present your answers is part of the marking.
5. Structure your code, include comments, and avoid unnecessary loops, conditions, and calculations. The style in which you write your code is also part of the marking.
6. The deadline for handing-in your assignment is **16:00 on Thursday, 26 March 2026**.

   Please upload exactly two files on Learn: a PDF with your report and a zip file containing all your codes.

Data

I have uploaded one zip file on Learn which contains eight data sets of different size from the OR Library. Please see

`https://people.brunel.ac.uk/~mastjjb/jeb/orlib/capinfo.html`

for an explanation of the format of the data sets.

1 Lagrangian Relaxation (100 marks)

In this assignment, we consider the capacitated warehouse location problem (CWLP). In this problem, we are given a set of clients, indexed as $I = \{1, \ldots, n\}$, who have demand for a generic product. Moreover, we are given a set of candidate locations, indexed as $J = \{1, \ldots, m\}$, where we can build new warehouses to serve the customers' demands. Each warehouse hereby has a capacity, which limits the amount of demand it can ship to customers.

The total amount of demand of customer $i \in I$ is denoted by b_i. The cost of serving all demand of customer i from a warehouse at location $j \in J$ is denoted as c_{ij}. The demand of a customer can either be served fully from a single warehouse or the customer can be served from multiple warehouses, in which case the shipping cost is proportional to the fraction of the customers' demand that is served from that warehouse.

Moreover, the cost of building a warehouse at location $j \in J$ incurs a fixed setup cost of f_j and the capacity of a warehouse built at j is denoted by K_j.

Defining the decision variables

$$\begin{aligned} x_{ij} &= \text{the fraction of demand of customer } i \in I \text{ that is served by a warehouse at location } j \in J \\ y_j &= \begin{cases} 1, & \text{if a warehouse is built at location } j \in J \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

the mixed-integer linear programming formulation for the CWLP is

$$\begin{aligned} \min \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} + \sum_{j \in J} f_j y_j & \\ \sum_{j \in J} x_{ij} = 1 & \qquad i \in I \\ \sum_{i \in I} b_i x_{ij} \leq K_j y_j & \qquad j \in J \\ x_{ij} \geq 0 & \qquad i \in I, j \in J \\ y_j \in \{0, 1\} & \qquad j \in J \end{aligned}$$

Questions

To solve the CWLP, derive and implement a Lagrangian relaxation and the corresponding subgradient optimization method by relaxing the customer allocation constraints.

Run your algorithm on each of the data sets provided and report its runtime, the objective value of the best found feasible solution, and the value of the best found lower bound (lower, as we are minimizing). Use the latter to assess the quality of the solution.

Hint: Using Lagrangian multipliers $\lambda = (\lambda_i)_{i \in I}$, the Lagrangian subproblem for this relaxation is given by

$$\begin{aligned} LS(\lambda) \;=\; \min \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} + \sum_{j \in J} f_j y_j + \sum_{i \in I} \lambda_i \left(1 - \sum_{j \in J} x_{ij} \right) & \\ \sum_{i \in I} b_i x_{ij} \leq K_j y_j & \qquad j \in J \\ 0 \leq x_{ij} \leq 1 & \qquad i \in I, j \in J \\ y_j \in \{0, 1\} & \qquad j \in J \end{aligned}$$

You have full freedom on how you approach this and how you set up your algorithm. Below, I give you a list of aspects that might be worthwhile considering. Please note that this list is not exhaustive, either because there is something I haven't thought about that might be interesting to explore or it's something I left out deliberately. And neither is this a mandatory check list. If you think some aspect doesn't make sense, then feel free to just ignore it.

1. What is the most efficient way to solve the Lagrangian subproblem for a given vector λ? Can you even solve it without needing a MIP solver?
2. What are good parameter settings for the subgradient optimization method and how to set its stopping criteria?
3. If an optimal solution of the Lagrangian subproblem is infeasible for the primal problem, i.e., the CWLP, is there an easy way to "repair" this solution, i.e., tweak the values of some of the decision variables so that the solution becomes feasible.
4. What's the best way to present your computational results?
5. How does your Lagrangian relaxation algorithm compare to solving the problem optimally with a solver, e.g., Xpress, CPLEX, or Gurobi.